

Math 104 – Real Analysis

Homework 2 — my solutions

Fall 2026 ■ Keshav Ramamurthy

Homework 2

Deadline: 11:59 pm September 13, 2026

Problem (1). Let S be a nonempty bounded subset of $\mathbb{R}$, and define the diameter of S by

$$\text{diam}(S) = \sup\{|s - t| : s, t \in S\}.$$

Prove that $\text{diam}(S) = \sup S - \inf S$.

Problem (2). Prove that for every real number $a > 0$ there is a unique natural number n such that

$$n - 1 \leq a < n.$$

(Your proof will need both the Archimedean Property and the Well-Ordering Principle for $\mathbb{N}$, together with an argument for uniqueness.)

Problem (3). Let

$$S = \{x \in \mathbb{R} : x^4 - 5x^2 + 4 < 0\}.$$

(a) Write S in interval notation.

(b) Find $\inf S$ and $\sup S$, and state clearly whether each is attained.

Problem (4). For each sequence below, determine whether it converges and, if so, find its limit. (You may use the limit theorems; no ε - N proof is required here, but you must justify divergence when it occurs, e.g. by exhibiting two subsequences with different limits.)

$$(a) a_n = \frac{2n^3 + n}{n^3 - 5}, \quad (b) b_n = (-1)^n \frac{n}{n + 1}.$$

Problem (5). Give rigorous ε - N proofs of the following limit.

$$\lim_{n \rightarrow \infty} \frac{3n^2 - 1}{2n^2 + n + 5} = \frac{3}{2}.$$

Solutions

Problem 1

Solution. We're given a nonempty bounded subset $\mathbb{R}$, and

$$\text{diam}(S) = \sup\{|s - t| : s, t \in S\}$$

Notice that by definition of $\sup$, $\text{diam}(S)$ is the least upper bound of the value of $|s - t|$ for two elements $s, t \in S$.

We notice from the Completeness Axiom on $\mathbb{R}$ (and thus on its non empty bounded subset), that $\sup(S)$ and $\inf(S)$ exist.

We first show that $\text{diam}(S) \leq \sup(S) - \inf(S)$.

Notice that for arbitrary s, t that $\inf(S) \leq s \leq \sup(S)$ and $\sup(S) \geq t \geq \inf(S)$ by definition, so subtracting these, $s - t \leq \sup(S) - \inf(S)$.

Since s and t are defined arbitrarily, their definitions can be flipped, yielding $t - s \leq \sup(S) - \inf(S) \rightarrow |s - t| \leq \sup(S) - \inf(S)$ thus $\text{diam}(S) \leq \sup(S) - \inf(S)$

Now we wish to show the other direction, namely $\text{diam}(S) \geq \sup(S) - \inf(S)$.

Since $\sup(S)$ exists, $\exists s \in S$ such that $s + \frac{\epsilon}{2} \geq \sup(S)$

Considering the opposite direction, $\exists t \in S$ such that $t - \frac{\epsilon}{2} \leq \inf(S)$ so rearranging, we have

$$s \geq \sup(S) - \frac{\epsilon}{2} \tag{1}$$

$$t \leq \inf(S) + \frac{\epsilon}{2} \tag{2}$$

$$\tag{3}$$

So $s - t \geq \sup(S) - \inf(S) - \epsilon$ Flipping t and s (where here,

$$\exists t \in S \text{ s.t. } t + \frac{\epsilon}{2} \geq \sup(S), \exists s \in S \text{ s.t. } s - \frac{\epsilon}{2} \geq \inf S$$

) just like above yields

$$\exists s, t \in S \text{ s.t. } t - s \geq \sup(S) - \inf(S) - \epsilon \iff |s - t| \geq \sup(S) - \inf(S) - \epsilon$$

This implies

$$\text{diam}(S) \geq \sup(S) - \inf(S) - \epsilon$$

So we've shown that

$$\sup(S) - \inf(S) - \epsilon \leq |s - t| \leq \text{diam}(S) \leq \sup(S) - \inf(S)$$

so we are done, since $\text{diam}(S)$ is bounded above by $\sup(S) - \inf(S)$ and epsilon can be arbitrarily

small, bounding it below by $\sup(S) - \inf(S)$, showing equality. $\square$

Problem 2

Solution. Given a real number $a > 0$, notice by the archimidean property, the set

$$W = \{k | k > a, k \in \mathbb{Z}^+\}$$

is nonempty, and by the well ordering principle (every non empty subset of $\mathbb{Z}^+$ has a unique minimum) so write $n = \min(W)$.

Thus, for any positive real a , there exists $n \in \mathbb{Z}^+$ s.t $n > a$

Claim $n - 1 \leq a$.

Proof. Notice by definition, n is the smallest integer in $\mathbb{Z}^+$ greater than a , meaning that $n - 1$ must be a or smaller, as it is not in set W . $\square$

Claim: n is unique

Proof. The uniqueness can be shown by contradiction.

Suppose there exists $n' \in W$ such that $n' \neq \min(W)$ and $a \in [n' - 1, n')$. (so the existence of another value in W that satisfies the same condition that holds for n .)

Notice that $n' \neq \min(W) \rightarrow n' > \min(W)$ and since we are working on integers, $n' \geq \min(W) + 1 \rightarrow n' - 1 \geq \min(W) > a$.

This yields a contradiction, as we originally had $a \in [n' - 1, n')$, yet we showed $n' - 1 > a$. $\square$

$\square$

Problem 3

Solution. We are given the set

$$S = \{x \in \mathbb{R} : x^4 - 5x^2 + 4 < 0\}$$

Notice that we can factor this expression, namely

$$x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4)$$

We investigate when this is zero, and we do casework on x^2 .

Case 1: $0 \leq x^2 \leq 1$ (the 0 is just from the trivial inequality) In this case, $x^2 - 1$ and $x^2 - 4$ are both less than or equal to 0, so the inequality is NOT satisfied. (either 0 or negative)

Case 2: $1 < x^2 < 4$ In this case, notice that $x^2 - 1$ is positive, but $x^2 - 4$ is negative, so the inequality is satisfied.

Case 3: $4 \leq x^2$ In this case, notice that $x^2 - 1$ is positive, and $x^2 - 4$ is greater than or equal to 0, so this expression will be greater than or equal to 0.

Thus, the only interval that works is $1 < x^2 < 4$ works, which yields $1 < x < 2$ and $-1 > x > -2$

So for part A, $S = (-2, -1) \cup (1, 2)$

For Part B:

$\inf(S)$ is -2 as every element in S is greater than -2 , and any element greater than -2 in $(-2, -1)$ can't be a lower bound, because if you consider $t \in (-2, -1)$, $\frac{t+(-2)}{2}$ is closer to -2 than t , a contradiction. (and if t is greater than or equal to -1), then any element in $(-2, -1)$ is smaller than t , so it can't be a lower bound.

Similarly, $\sup(S)$ is 2 because any element in $(-2, -1) \cup (1, 2)$ is smaller than 2 . We can do the same argument. If $t \in (1, 2)$, $\frac{t+2}{2}$ is closer to 2 than t , and contained in $(1, 2)$, and again if $t \in (-2, -1)$, then obviously any element in $(1, 2)$ is greater than t , so t can't be an upper bound. $\square$

Problem 4

Solution. Scratch Work:

we consider $|a_n - 2| < \epsilon$. We get

$$|a_n - 2| < \epsilon \frac{2n^3 + n}{n^3 - 5} - \frac{2(n^3 - 5)}{n^3 - 5} < \epsilon \quad (4)$$

$$\frac{n + 10}{n^3 - 5} < \epsilon \quad (5)$$

$$\frac{n + 10}{n^3 - 5} > \frac{n + 10}{n^3} > \frac{1}{n^2} \quad (6)$$

$$(7)$$

Claim. For large enough n ,

$$\left| \frac{2n^3 + n}{n^3 - 5} - 2 \right| < \frac{2}{n^2} < \epsilon$$

Proof. Let's cross multiply:

$$\frac{n + 10}{n^3 - 5} < \frac{2}{n^2}$$

Then we get

$$n^3 + 10n^2 < 2n^3 - 10 \rightarrow 10n^2 + 10 \leq 11n^2 \leq n^3 < 2n^3$$

which is true for any n greater than, for example $n = 100$. (LHS: 100,010 RHS: 2000000). $\square$

Thus, we solve $\frac{2}{n^2} < \epsilon \rightarrow \frac{2}{\epsilon} < n^2$ so for positive n , $n > \sqrt{\frac{2}{\epsilon}}$ Thus, we claim the answer holds for

$$N = \max \left(100, \left\lceil \frac{2}{\epsilon} \right\rceil + 5 \right)$$

, and it holds as is shown in the scratch work.

Part b) We claim that this sequence diverges.

This is easily shown by considering what a_n converges to when n is even or odd. When n is even, suppose $n = 2k$ for $k \in \mathbb{Z}^+$, we have

$$\lim_{k \rightarrow \infty} b_{2k} = (-1)^{2k} \frac{2k}{2k+1} = 1.$$

Yet, when considering:

$$\lim_{k \rightarrow \infty} b_{2k+1} = (-1)^{2k+1} \frac{2k+1}{2k+2} = -1$$

Since the even and odd subsequences converge to different values, this sequence is divergent. $\square$

Problem 5

Solution. Scratch Work: Consider $\left| \frac{3n^2-1}{2n^2+n+5} - \frac{3}{2} \right| < \epsilon$

Then, we can distribute to get

$$\left| \frac{6n^2 - 1 - 3(2n^2 + n + 5)}{(2n^2 + n + 5)(2)} \right| < \epsilon$$

which reduces to

$$\frac{3n + 16}{(2n^2 + n + 5)(2)} < \epsilon$$

Notice that

$$\frac{3n + 16}{(2n^2 + n + 5)(2)} > \frac{3n}{(2n^2 + n + 5)(2)} > \frac{3n}{4n^2} > \frac{3}{4n}$$

We wish to see and find a constant $c \cdot \frac{3}{4n}$ that is greater than

$$\frac{3n + 16}{(2n^2 + n + 5)(2)} < c \frac{3}{4n}$$

Consider $c = 2$. Then, cross multiplying:

$$(4n)(3n + 16) < 6(2n^2 + n + 5)(2)$$

So

$$12n^2 + 64n < 12n^2 + 12n + 60$$

for $n > 64$ Thus, we claim that $N = \max \left(65, \left\lceil \frac{3}{2\epsilon} \right\rceil + 5 \right)$ $\square$